

A provenance-sealed anomaly-score candidate catalogue of DESI DR1 spectra: selection, validation, and a descriptive taxonomy

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We publish a machine-flagged candidate catalogue of spectroscopically unusual objects in Data Release 1 of the Dark Energy Spectroscopic Instrument (DESI), together with the complete provenance chain needed to reproduce it. An archived convolutional autoencoder, bound by SHA-256 to a sealed run contract, assigns a reconstruction-error anomaly score S to every science-target spectrum in the public `iron` release; 28,425,963 raw rows deduplicate to 27,547,223 unique `TARGETIDs`, of which 21,793,550 pass the science-target provenance gate. We show that a score cut alone does not select astrophysical objects: above $S = 3$, more than 99.5% of raw fibres in every score bin are sky or non-science fibres, and that fraction *rises* with score. Applying the provenance gate (`OBJTYPE = TGT`, `COADD_FIBERSTATUS = 0`, `TARGETID > 0`) before any score cut, and a pre-declared threshold rule, yields a released catalogue of 1,244 objects — the top 5.7×10^{-5} of the science-target population. We validate the catalogue against a published contract of sixteen machine-checkable requirements, and report three results that constrain its interpretation: the anomaly score is uncorrelated with exposure quality and source brightness ($|\rho_s| < 0.1$), so the tail is not a faintness artefact; the score is driven almost entirely by the blue spectrograph arm ($\rho_s = +0.53$ against the b -camera residual, versus -0.04 and -0.14 for r and z); and the catalogue’s descriptive taxonomy of 25 clusters and 8 families, built over the 675 objects with no SIMBAD or NED counterpart, carries no material structure in the model’s own latent space and must be read as a stratification of the candidate list rather than as physical classes. A known-object recovery benchmark against five reference classes recovers no class at the pre-declared bar, so this work is reported as a validated data release and not as a discovery. We name 32 follow-up targets in five rules-based tiers and run the cheapest of the named tests here: of four anomaly-selected DESI-pipeline $z \geq 4$ quasar candidates with no catalogued counterpart, the released Legacy Survey photometry already refutes one and supports one, and cannot decide the remaining two because the published schema omits flux inverse variances — one of four release defects we record openly rather than hide. We close with the open tests, led by the absent selection function, that any population-level use of this catalogue must clear first.

I. INTRODUCTION

Wide-field spectroscopic surveys produce far more spectra than can be inspected. Unsupervised outlier detection is the standard response [1], and it works: a model trained to reconstruct typical spectra flags the ones it cannot. The approach has been carried into DESI itself, both on the Bright Galaxy Survey [2] and with variational autoencoders on galaxy spectra [3].

What such a search usually does not publish is the part that decides whether its output means anything — the exact model, the exact code, the exact parent catalogue, the selection rule, the reason the rule was chosen, and the tests the resulting list did *not* pass. Without those, a candidate list is unfalsifiable: a reader cannot tell a genuine spectral outlier from a fibre placed on blank sky, and cannot reproduce either.

This paper publishes that part alongside the catalogue. Our contributions are:

1. **A sealed provenance chain** (Sec. IIB): the SHA-256 of the parent `zcatalog`, of the archived model, of the inference code, and of every intermediate product,

bound under one run contract and re-verified end to end at publication time.

2. **A demonstration that the provenance gate, not the score cut, is what makes such a sample astrophysical** (Sec. IIIB). Above $S = 3$ the raw fibre population is more than 99.5% sky or non-science in every score bin, and the contamination *worsens* with score. This is a property of the substrate, not of our model, and it applies to any reconstruction-error search run on public coadds.
3. **A published validation contract** of sixteen machine-checkable requirements (Sec. V), with its checker and raw output released, including two defects the release carries openly.
4. **Three measurements of what the score responds to** (Secs. VB–VIC): it is not an exposure-quality proxy, it is driven by the blue arm, and the taxonomy it supports is descriptive rather than physical.
5. **A negative recovery benchmark** (Sec. VD): against five published reference classes, no class is recovered at the pre-declared bar.
6. **A named, rules-based follow-up set** (Sec. VII), with the cheapest test in the set — a Lyman-break

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consistency check on public imaging — executed here rather than deferred.

The framing follows from the benchmark. A catalogue whose pre-declared success condition is unmet is a data release with a validated selection, not a discovery paper, and we report it as one throughout. Section II describes the data and the provenance chain, Sec. III the model and the selection rule, Sec. IV the catalogue itself, Sec. V the validation contract and what it establishes, Sec. VI the descriptive taxonomy, Sec. VII the named follow-up set and the open tests, Sec. VIII what this catalogue does and does not bear on cosmologically, Sec. IX the reproduction path, and Sec. X what this catalogue is not.

II. DATA

A. DESI DR1

We use Data Release 1 of DESI [4, 5]: the public `iron` spectroscopic reduction, its redshift catalogue `zall-pix-iron.fits`, and the per-(survey, program, healpix) coadd spectra, streamed and scored in place rather than bulk-downloaded. Redshifts, spectral classifications and quality flags are the pipeline’s own [6, 7].

The scan covered a 36,634-group locator inventory, receipt-verified group by group, and processed 28,425,963 raw rows. Deduplication by last `TARGETID` occurrence in lexical shard-then-row order removed 878,740 duplicate rows and left 27,547,223 unique `TARGETIDs`. Of those, 21,793,550 satisfy the science-target provenance gate defined in Sec. III B; that number, not the full 27.5 million, is the parent population of this catalogue, and every selected fraction quoted below uses it.

B. Provenance chain

Every stage binds the exact SHA-256 of its input and output under a single sealed run contract (generation `clean-rerun-6699d09ff886`). The thirteen hash-bound artifacts are listed in Table I: the parent `zcatalog`, the run contract itself, the archived model and inference code, and the nine released data products.

All eight links of the manifest chain — sample → enrichment → cross-match → taxonomy, each stage naming its input hash and its output hash — are verified in the validation contract (check V3, Table VI), and all nine released artifacts re-hash to the values in the landing receipt (check V2).

III. METHOD

A. The archived model and its calibration

Spectra are scored by an archived autoencoder frozen for this run. The input is a per-spectrum median-absolute-flux-normalised 496-bin resampling; the encoder compresses it to 128 latent dimensions, and the score is a standardised reconstruction-error statistic,

$$S = \frac{\overline{\text{MSE}} - \mu_{\text{MSE}}}{\sigma_{\text{MSE}}}, \quad (1)$$

with $\overline{\text{MSE}}$ the per-spectrum mean squared reconstruction error and $(\mu_{\text{MSE}}, \sigma_{\text{MSE}}) = (0.8771, 1.3605)$ sealed before the scan. A high S therefore means a spectrum unlike the shapes the model learned to reproduce; it is a distance in the model’s own reconstruction metric and carries no physical normalisation.

The calibration constants were fit on a two-stage probability-proportional-to-size cluster sample of 200 groups and 20,000 held-out spectra, with a further 20,000 reserved for a stability check. The check requires $|\mu_{\text{MSE}}^{\text{val}} - \mu_{\text{MSE}}| \leq 5 \sigma_{\text{MSE}}/\sqrt{n}$; the observed deviation was 0.00543 against a bound of 0.0481, and it passed. The constants and the model weights were then frozen and hash-bound before any spectrum of the survey was scored, so the score of a given spectrum is fixed by the contract rather than by the sample it is scored with.

Across the full scan the score distribution is sharply peaked and long-tailed: over all fibres the median is -0.56 , the 99th percentile is 3.08, and the maximum observed is 24.42. Over science targets alone the distribution is tighter — median -0.60 , 99th percentile 0.96, maximum 11.33. The released threshold of $S > 3$ therefore sits far above the science-target 99th percentile, and the released catalogue is the top 5.7×10^{-5} of the science-target population.

B. The science-target provenance gate

A row enters the candidate pool only if `OBJTYPE = TGT` and `COADD_FIBERSTATUS = 0` and `TARGETID > 0`, the last asserted rather than merely filtered. This gate is applied *before* any score cut and is enforced by a committed, re-runnable checker, `check_sample_provenance.py` (full path in the reproducibility manifest).

The gate is not a formality, and the quantitative reason is the single most transferable result in this paper. Figure 1 and Table II give the fraction of raw fibres that are sky or non-science as a function of score: it exceeds 99.5% in every bin above $S = 3$ and *increases* monotonically with score, from 99.53% in [3, 4) to 99.97% in [8, 10); the sparsely populated top bin (337 fibres, one science target) sits at 99.70%. A naive top-score selection from this scan returns instrument artefacts. The effect is a property of the public coadd substrate — sky fibres

TABLE I. The hash-bound provenance chain. Every artifact is re-hashed at publication time and compared against the landing receipt; all thirteen match (checks V2 and V3).

Artifact	SHA-256 (first 16 hex)
DESI DR1 zcatalog zall-pix-iron.fits	2d95ad99361039b5...
sealed run contract (clean-rerun-6699d09ff886)	6699d09ff886f74d...
archived model best_model.47k.pt	f5266ba48f476bca...
inference code	3e7efb243fa5cc4e...
flagship_crossmatch_v2_matched.parquet	4a718d1a7ad253d2...
flagship_crossmatch_v2_unmatched.parquet	c0a6b57bd672f81b...
flagship_sample_v2.parquet	d6d43dfa04d6a8b2...
flagship_sample_v2_enriched.parquet	c3b176ff2d355a42...
flagship_taxonomy_v2.json	1420388b59f37278...
flagship_wise_v2.parquet	1b42125d5e62830c...
science_target_summary.json	439f886bd927625d...
sky_fraction_by_score.json	b2741b18dd543157...
threshold_choice.json	e4244bbab5779c87...

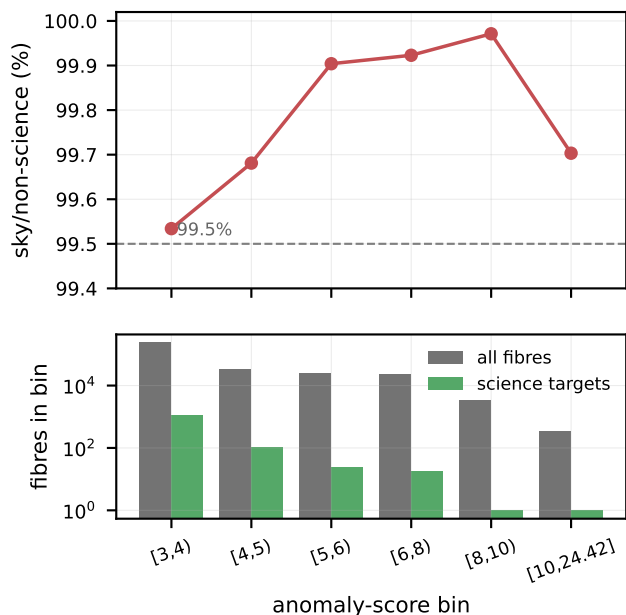


FIG. 1. *Top*: sky-or-non-science fibre fraction per score bin, before the provenance gate; note the suppressed zero. The fraction exceeds 99.5% in every bin and rises monotonically to 99.97% in [8, 10). *Bottom*: the bin populations behind it. Redrawn from the released `sky_fraction_by_score.json`.

are reconstructed worst precisely because they contain no source — so any reconstruction-error search on DESI coadds that omits a provenance gate will report the same population, whatever its model.

TABLE II. Per-bin fibre census before the provenance gate. “Sky/non-science” is NOT (OBJTYPE = TGT AND COADD_FIBERSTATUS = 0), with TARGETID ≤ 0 counted as sky/non-science.

Score bin	All fibres	Science targets	Sky/non-science
[3, 4)	234,365	1,092	99.53%
[4, 5)	33,865	108	99.68%
[5, 6)	25,008	24	99.90%
[6, 8)	23,370	18	99.92%
[8, 10)	3,473	1	99.97%
[10, 24.42]	337	1	99.70%

C. The threshold rule

The release threshold follows a rule fixed before the science-only counts were inspected: from the grid {3, 4, 5, 6, 8, 10}, take the largest threshold whose science-only count is ≥ 300; if that count exceeds 1,500, step to the next grid point unless doing so drops the count below 300. The rule selects $S > 3$ with $n = 1,244$, reproduced independently from the counts rather than read from the stored choice in check V4. Table III gives the raw and science-target counts on the ladder; Fig. 2 plots both survival curves alongside the released sample’s score distribution. The two curves diverge by more than two orders of magnitude, which is the same statement as Fig. 1 read cumulatively.

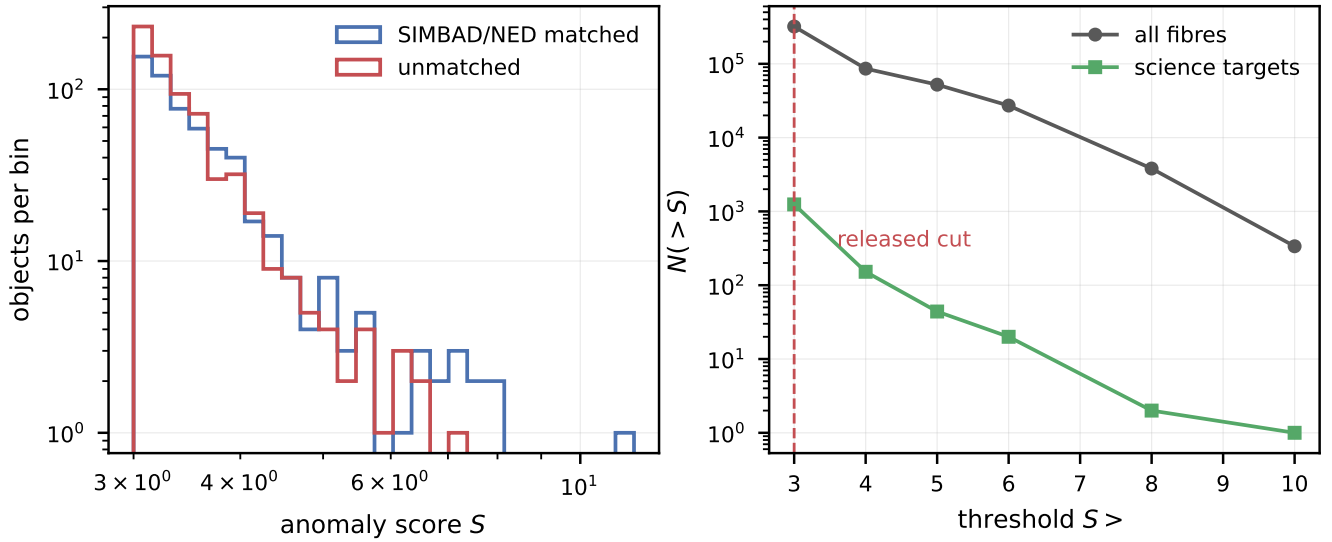


FIG. 2. *Left*: anomaly-score distribution of the 1,244 released objects, split by whether a SIMBAD or NED counterpart exists. *Right*: survival counts $N(> S)$ on the pre-declared threshold grid for all fibres and for science targets; the released cut is marked. The gap between the two curves is the sky-fibre population removed by the provenance gate.

TABLE III. Threshold ladder: all fibres versus science targets.

$S >$	All fibres	Science targets	Science fraction
3	320,418	1,244	0.39%
4	86,053	152	0.18%
5	52,188	44	0.08%
6	27,180	20	0.07%
8	3,810	2	0.05%
10	337	1	0.30%

IV. THE CATALOGUE

A. Contents

The released catalogue contains 1,244 objects. Table IV lists the published columns by group: identification and sky position, the anomaly score with its per-camera residual decomposition, the 128 latent activations, the DESI pipeline redshift solution, exposure-quality diagnostics, Legacy Survey photometry [14], the AllWISE and SIMBAD/NED cross-match results, and the taxonomy labels of Sec. VI.

Scores range from 3.0002 to 11.3316, with a median of 3.28 and a 90th percentile of 4.08: the catalogue is dominated by objects just above threshold, as any tail-selected sample is.

B. Redshift reliability

Of the 1,244, 502 (40.4%) carry a reliable pipeline redshift ($ZWARN = 0$); the remaining 742 carry $ZWARN = 4$ (`SMALL_DELTA_CHI2`) and their redshifts must not be used as measurements [7]. Every redshift-dependent statement in this paper is restricted to the $ZWARN = 0$ subset, which has median $z = 1.14$ and splits 424 `GALAXY` / 73 `QSO` / 5 `STAR`; 70 objects sit at $z > 2$, 36 at $z \geq 4$ and 16 at $z \geq 6$, with a maximum of $z = 6.75$. Pipeline classifications over the whole sample are 1,153 `GALAXY`, 86 `QSO` and 5 `STAR`. Figure 3 shows the reliable-redshift distribution and its relation to the score; there is no visible trend of score with redshift, and the high-redshift objects are not the high-score ones.

C. Photometry and morphology

Legacy Survey photometry is carried for every object. 441 are typed `PSF` (point-like) and 316 `REX`, with 406 rows carrying an empty `MORPHTYPE` — a gap in the photometric join that the deposit must close. Only 6 objects have a Gaia DR3 counterpart with a measured G magnitude [15] and only 5 a non-zero parallax, so the sample is, with a handful of exceptions, fainter than Gaia’s limit; none of the 675 unmatched objects has a Gaia counterpart at all. The median coadd exposure time is 1,129 s. AllWISE counterparts exist for 74 objects, with median $W1 - W2 = 0.51$.

The released schema does *not* carry flux inverse variances. That omission is consequential — it prevents a reader from forming a detection significance for any pho-

TABLE IV. Released column schema, by group.

Group	N_{col}	Columns
identification	6	targetid, target_ra, target_dec, survey, program, healpix
anomaly	7	anomaly_score, mean_mse, rB, rR, rZ, worst_band, peak_residual_wavelength
latent	128	latent_000–latent_127
DESI pipeline	4	z, zwarn, spectype, deltachi2
exposure	9	coadd_numexp, coadd_exptime, median_coadd_snr_b, median_coadd_snr_r, median_coadd_snr_z, tsnr2_qso, tsnr2_elg, tsnr2_lrg, tsnr2_bgs
Legacy Survey photometry	11	flux_g, flux_r, flux_z, flux_w1, flux_w2, morphtype, sersic, shape_r, ebv, parallax, gaia_phot_g_mean_mag
AllWISE cross-match	5	w1, w2, w1_w2, match_separation_arcsec, match_flag
SIMBAD/NED cross-match	6	simbad_found, simbad_main_id, simbad_otype, ned_found, ned_name, ned_type
taxonomy	4	cluster_id, family_id, family_descriptor, is_core_member

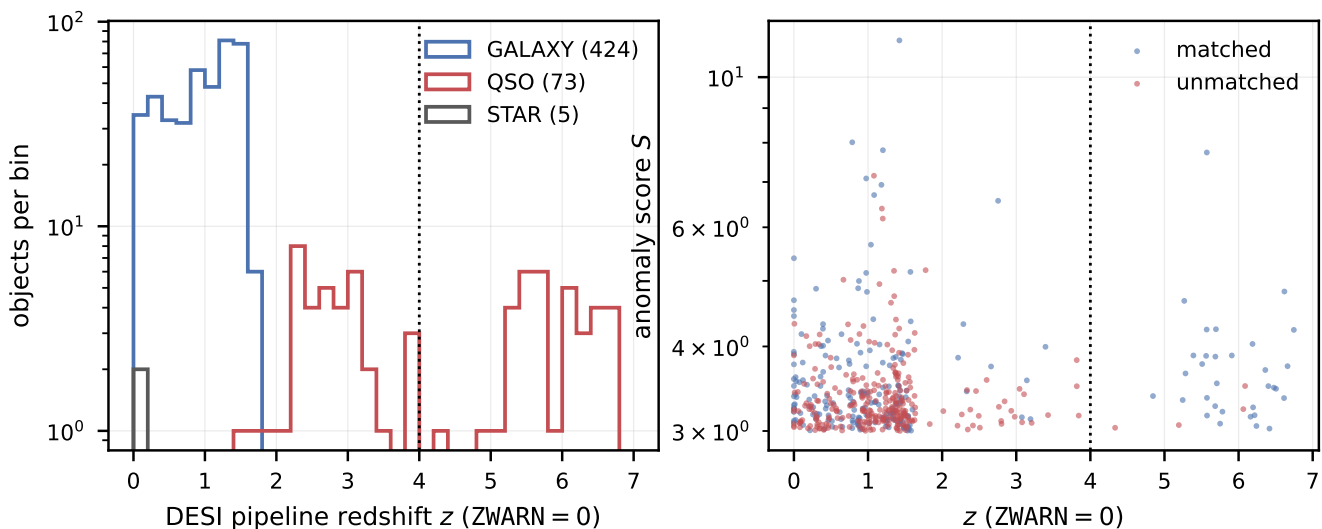


FIG. 3. *Left*: DESI pipeline redshifts of the 502 objects with ZWARN = 0, by pipeline classification; the $z = 4$ line marks the FT-A tier boundary. *Right*: score against redshift for the same subset. High score and high redshift are not the same population.

tometric statement, including the Lyman-break test of Sec. VII B — and it is recorded here as a release defect to be repaired in the public deposit.

D. External counterparts

SIMBAD [8] and NED cross-matching at $3''$ identifies counterparts for 569 objects (45.7%): 562 via NED, 38 via SIMBAD, 31 via both. That leaves 675 (54.3%) unmatched. Table V breaks the matches down by counterpart type. They are overwhelmingly ordinary: 455 of the NED matches are typed G (galaxy) and 52 IrS (infrared source), with 31 QSO; the SIMBAD side, which matches far fewer objects, is relatively richer in quasars (19 of 38).

Unmatched does not mean new. The unmatched set includes objects simply outside other surveys' footprints or

TABLE V. Counterpart types of the 569 matched objects, by service. NED types dominate by count; SIMBAD matches fewer objects but a larger quasar fraction. Only the eight most common types per service are listed.

NED type	N	SIMBAD type	N
G	455	QSO	19
IrS	52	G	12
QSO	31	*	4
*	14	PN	1
VisS	4	rG	1
UvS	3	WD?	1
RadioS	2		
XrayS	1		

below their detection limits, and the Gaia statistic above shows the population is faint. The point is reinforced by the extreme tail: the three highest-scoring objects in the whole sample ($S = 11.33, 8.01, 7.80$) are all *matched*, i.e. already catalogued. The score’s strongest responses are to known objects, which is a mild positive signal for the method and a reminder that the unmatched subset is not the interesting tail by construction.

Figure 4 shows the sky distribution of both subsets. The catalogue traces the DR1 footprint and its tiling pattern; neither subset is concentrated anywhere, and the matched and unmatched objects are interleaved.

V. VALIDATION

A. The validation contract

Table VI lists the requirements this release is held to. Each is machine-checkable against the published files; the checker and its raw output are released with the catalogue. Requirements cover provenance and hash binding (V1–V3b), the selection rule (V4, V7, V9), internal consistency of the released tables (V5, V6, V8), three tests of what the score measures (V10–V12), and three defects the release must state rather than hide (DEFECT-1 to DEFECT-3).

Thirteen requirements pass. V11 and DEFECT-3 are REPORTED quantities rather than thresholds. V12 returns STRUCTURED-BUT-WEAK, and we take the weaker reading throughout (Sec. VIC). DEFECT-1 and DEFECT-2 FAIL by design: they exist to force the release to carry its own known defects (Sec. VE).

B. The score is not an exposure-quality proxy

The first thing to rule out is that the score measures how badly a spectrum was measured rather than what it contains. Within the science-target sample the anomaly score is essentially uncorrelated with every available measure of exposure quality: Spearman $\rho_s = +0.058$ against median coadd signal-to-noise, -0.076 against coadd exposure time, -0.093 and -0.096 against $\text{TSNR2}_{\text{LRG}}$ and $\text{TSNR2}_{\text{QSO}}$, and -0.016 against r -band flux (Fig. 5). At $n = 1,244$ four of these are formally significant and all are negligible in size; the significant ones are negative, i.e. longer exposures score marginally *lower*.

The claim this supports is narrow and negative: the $S > 3$ tail is not explained by faintness or short exposure. It is not evidence that the tail is astrophysically interesting. A referee should read V10 as closing one explanation, not as opening another.

C. The score is driven by the blue arm

DESI spectrographs record each fibre in three cameras, and the released catalogue carries the reconstruction residual of each separately. Decomposing by camera gives $\rho_s(S, r_B) = +0.527$ ($p \sim 10^{-89}$), against -0.037 ($p = 0.19$) for r_R and -0.142 ($p \sim 5 \times 10^{-7}$) for r_Z . The median residuals are correspondingly lopsided: $r_B = 2.53$ versus $r_R = 1.03$ and $r_Z = 0.94$. Figure 6 shows the three relations. The b -camera panel is a near-monotonic locus; the other two are scatter.

Whatever the model responds to lives in the b camera. Two readings survive this measurement, and the data here do not separate them: blue-end sky-subtraction and calibration residuals, which are a known difficulty at the blue limit of fibre spectrographs, or genuine spectral behaviour at rest frame wavelengths that land in the b camera for this redshift distribution. This is the single most consequential open question about the catalogue, because the first reading would make much of the list instrumental. It is also cheap to test on public data, and we name that test as OT-3 in Sec. VIIC rather than speculating here.

D. Known-object recovery

A candidate list can be tested against classes of genuinely unusual objects that are already published: if the score is finding unusual astrophysics, it should preferentially recover them. We cross-matched the released sample at $1.5''$ against eleven candidate reference classes, of which five had usable positions (four lacked identifiable coordinate columns, one had no known catalogue identifier, and one timed out). The bar was declared in advance: at least one class recovered at $> 10\times$ enrichment with ≥ 5 matches.

Table VII gives the outcome. One broad-absorption-line quasar [17] matches, at $4.2\times$ the base rate; RomabZCAT blazars [18], cataclysmic variables [19], Lyman- α emitters [20] and superluminous-supernova hosts [21] return nothing. *No reference class is recovered at the pre-declared bar*, and there is no closed-loop candidate class.

Two caveats sharpen rather than soften that conclusion. First, the released benchmark computes its base rate as 1,244 over all 27,547,223 unique TARGETIDs; the reference objects are science targets, so the fairer denominator is the 21,793,550 science-target parent, which raises the base rate and *lowers* the BAL enrichment from $4.2\times$ to $3.3\times$. Second, the reference footprints are small compared with the parent survey, so the test has little power for rare classes: the 95% upper limit on the recovery of superluminous-supernova hosts is 12.5%, which excludes almost nothing. The honest summary is that the benchmark rules out a strong recovery signal and is consistent with anything weaker.

This is the result that makes the present work a data

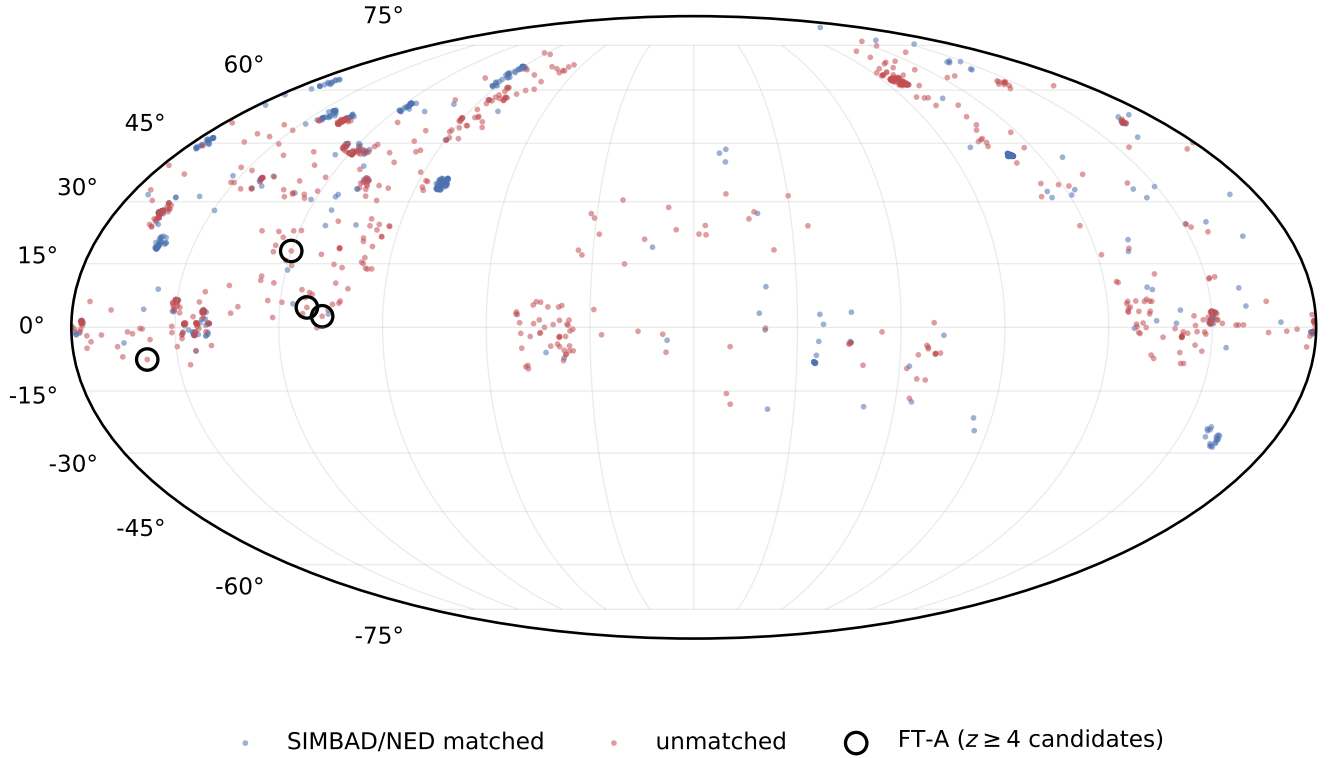


FIG. 4. Sky distribution of the released catalogue in equatorial coordinates, split by SIMBAD/NED counterpart status, with the four FT-A high-redshift candidates circled. The pattern traces DR1 tiling, not any property of the score.

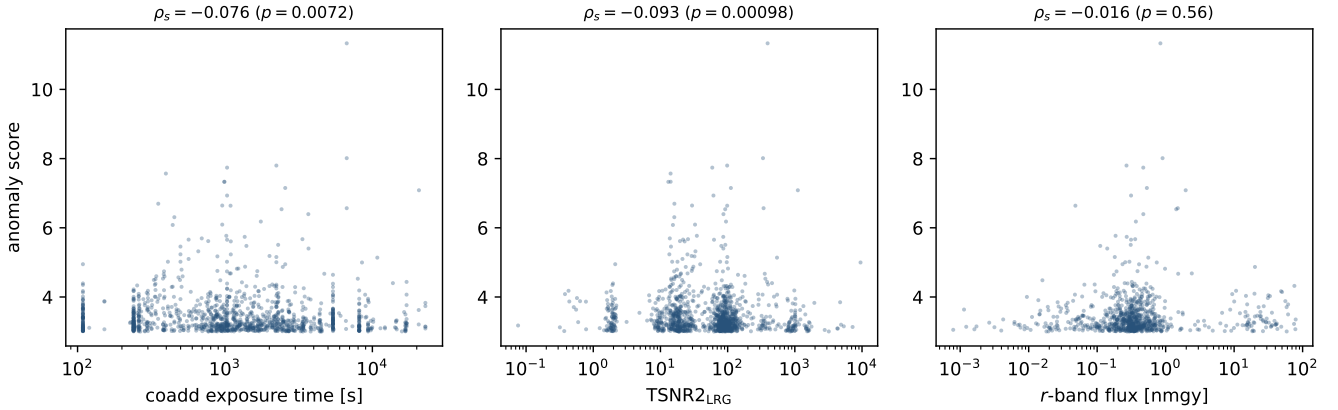


FIG. 5. Anomaly score against exposure time, $\text{TSNR2}_{\text{LRG}}$, and r -band flux for the 1,244 released objects, with Spearman coefficients.

release with a validated selection rather than a discovery paper, and we state it as such.

E. Known defects

Two defects in the released tables are recorded openly and must be repaired before public deposit. The enriched table's `z`, `ZWARN`, `SPECTYPE` and `SUBTYPE` columns are un-

populated, so a reader taking that file alone would read 1,244 null redshifts; the authoritative values are present in the sample table and are re-joined for every number in this paper, and the repair table is released alongside it. The `residual_kurtosis` column is empty; it is used by no result here and should be either populated or dropped from the published schema. Two further schema gaps are recorded in Sec. IV: the absent flux inverse variances, and the absent `OBJTYPE` and `COADD_FIBERSTATUS` columns,

TABLE VI. Validation contract. Outcomes are re-run against the released artifacts; REPORTED marks a quantity the release must state rather than a threshold it must clear.

ID	Requirement	Outcome
V1	gates/check_sample_provenance.py exits 0 on the released sample tables (TARGETID > 0; OBJTYPE/FIBERSTATUS where carried)	PASS
V2	every released artifact’s SHA-256 equals the value in the landing receipt (PHASE3_V2_LANDING_2026-09-03.md)	PASS
V3	the manifest chain closes end to end: sample $\rightarrow$ enrichment $\rightarrow$ cross-match $\rightarrow$ taxonomy, each stage binding the exact SHA-256 of its input and output, under one run contract, with the model / inference-code / zcatalog hashes recorded	PASS
V3b	the enrichment stage’s own MSE reproduction cross-check passed for every released row	PASS
V4	replaying the pre-declared threshold rule (largest grid point with science-only count ≥ 300 , step up if $> 1,500$) on the science-target counts selects the released threshold	PASS
V5	the cross-match output partitions the sample exactly: matched + unmatched = sample, disjoint, no duplicate TARGETIDs, and the enriched table covers the same TARGETIDs	PASS
V6	the taxonomy covers exactly the SIMBAD/NED-unmatched subset; 25 clusters roll up into 8 families; per-family counts sum to the unmatched count and match the per-object labels	PASS
V7	every released object exceeds the stated score threshold ($S > 3$)	PASS
V8	the AllWISE table covers every released object and its match count equals the released figure (74)	PASS
V9	the released sky-fraction-by-score validation supports the stated claim that every score bin above 3 is $> 99.5\%$ sky-or-nonscience before the provenance gate	PASS
V10	within the science-target sample the anomaly score is not a restatement of exposure quality or source brightness ($ \text{Spearman } \rho < 0.2$ against S/N, exposure time, TSNR2 and r-band flux)	PASS
V11	the per-band residual decomposition identifies which camera arm drives the score (reported, not gated)	REPORTED
V12	the released taxonomy’s family partition is tested for structure in the 128-dimensional latent space it was NOT built from (silhouette vs a label-permutation null, 50 draws)	STRUCTURED-BUT-WEAK
DEFECT-1	the enriched table’s zcatalog columns (z, zwarn, spectype, subtype) are populated	FAIL
DEFECT-2	every derived residual-diagnostic column in the enriched table carries values	FAIL
DEFECT-3	the fraction of released objects carrying a DESI-reliable redshift (ZWARN == 0) is stated with the catalogue	REPORTED

whose absence means the provenance gate cannot re-verify its own first two conditions from the published file alone. The deposit must carry all four.

up by descriptor identity into 8 families (Table VIII, Fig. 7). Family sizes sum to 675 exactly, and the per-family counts agree with the per-object labels (check V6).

VI. DESCRIPTIVE TAXONOMY

A. Construction

The 675 unmatched objects are grouped by PCA $\rightarrow$ UMAP [9] $\rightarrow$ HDBSCAN [10] on (S, α, δ) , with k -nearest-neighbour noise reassignment, giving 25 clusters that roll

B. What the families are

Because the clustering features are the score and the sky position, the families are in practice a score-tier $\times$ survey/programme stratification. Table IX makes this explicit: 70.4% of objects lie in their family’s dominant survey/programme cell, and no family is a compact sky

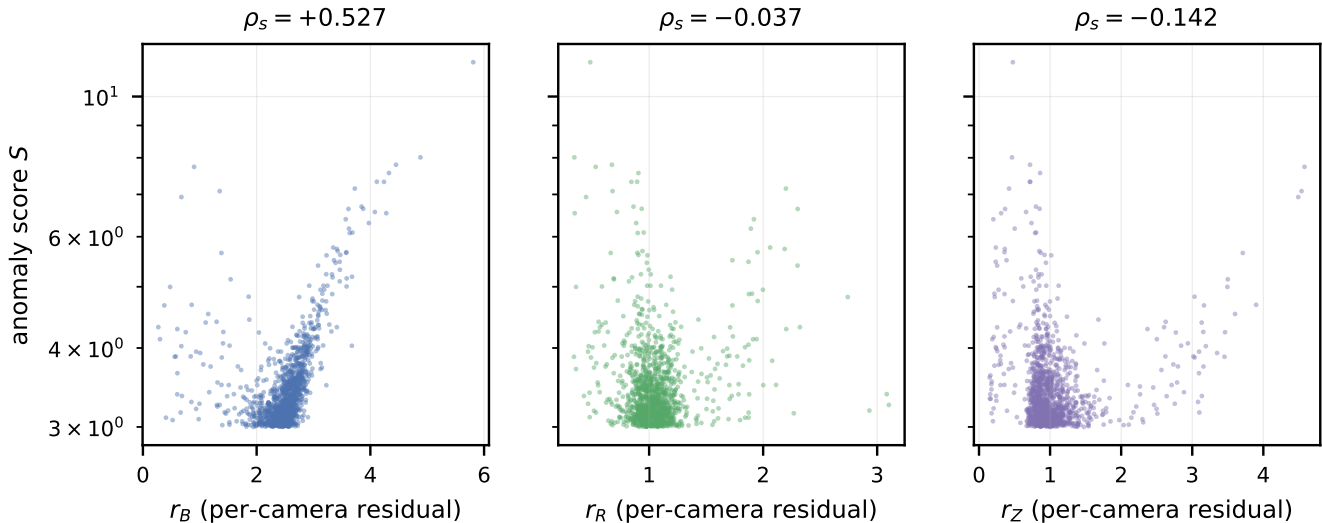


FIG. 6. Anomaly score against the per-camera reconstruction residual for the 1,244 released objects. The score tracks the b -camera residual and is uncorrelated with the other two.

TABLE VII. Known-object recovery benchmark, $1.5''$ positional match. N_{ref} counts reference objects inside the DR1 footprint; recovery is the matched fraction of those, with a 95% binomial interval; enrichment is recovery over the catalogue’s base rate in the parent.

Reference class	N_{ref}	N_{match}	Recovery (95% CI)	Enrichment
Broad absorption line (BAL) quasars	5,285	1	0.019% [0.003, 0.107]	4.2×
Roma-BZCAT blazars (5th edition)	2,060	0	0.000% [0.000, 0.186]	0.0×
Cataclysmic variables / white-dwarf binaries	580	0	0.000% [0.000, 0.658]	0.0×
Lyman- α emitters (LAEs)	84	0	0.000% [0.000, 4.373]	0.0×
Superluminous supernova (SLSN) host galaxies	27	0	0.000% [0.000, 12.456]	0.0×

region — the most concentrated still spans 51.7° in right ascension.

C. The latent-space test

A stratification built from the score and the sky is not, on its face, a claim about spectra. So we tested the partition in the space the model actually encodes spectra into, and which the clustering never saw: its silhouette over the 128 latent dimensions is -0.0162 , against a 50-draw label-permutation null of -0.0244 ± 0.0039 whose largest draw is -0.0184 (check V12, Fig. 8). The observed value exceeds every permutation draw, so the empirical $p < 1/50$: the partition is distinguishable from a random relabelling.

But the absolute silhouette is *negative*. A negative silhouette means the typical object is, on average, closer to some other family’s members than to its own. The families are therefore not separated clusters in latent space, and the correct reading is the weaker one: they are descriptive strata of the candidate list that carry a small amount of latent-space information, not physical classes.

They are labelled with descriptors only, and no family is given a physical class name. Re-clustering on the latents themselves is the obvious next version (OT-2).

D. Two families worth starting with

Two families stand out for follow-up purposes. Family 3 (61 objects, a single cluster) is the extreme-score family, with median $S = 4.38$, a 95th percentile of 6.31, the largest median blue residual of any family ($r_B = 2.98$), and the highest-scoring unmatched object in the catalogue ($S = 7.15$). Family 6 (36 objects) has the highest reliable-redshift fraction (27/36, 75%) at an elevated median score of 3.62 and no object above $z = 2$: well-measured, low-redshift, high-score, and therefore the family in which an instrumental or template explanation is most directly testable. Family 7 is its mirror — the same score tier with the worst redshift reliability (7/36, 19%), no quasar and no AllWISE match — and any follow-up there starts with re-reduction rather than with physics.

The families also inherit the limits of the cross-match.

TABLE VIII. Per-family evidence over the 675 SIMBAD/NED-unmatched objects. N_{cl} is the number of source clusters, r_B the median b -camera residual, N_{Z0} the number with $\text{ZWARN} = 0$, $\bar{z}_{Z0}$ their median redshift, and N_W the number with an AllWISE counterpart.

Family	Tier	N	N_{cl}	Survey/prog.	Score med. (5–95%)	r_B	N_{Z0}	$\bar{z}_{Z0}$	$N_{z>2}$	N_{QSO}	N_W
0	low	302	12	main/dark	3.14 (3.02–3.38)	2.42	144	1.15	13	18	0
1	elevated	87	4	main/dark	3.35 (3.12–3.78)	2.53	47	1.20	8	10	1
2	elevated	71	2	sv3/bright	3.53 (3.35–4.06)	2.60	18	1.36	2	4	2
3	extreme	61	1	sv3/bright	4.38 (3.55–6.31)	2.98	19	1.19	1	2	1
4	low	44	2	sv3/bright	3.10 (3.01–3.23)	2.42	12	1.36	3	3	1
5	low	38	2	sv1/other	3.12 (3.01–3.28)	2.20	15	1.01	2	4	1
6	high	36	1	main/dark	3.62 (3.44–4.31)	2.81	27	1.09	0	1	0
7	high	36	1	sv3/bright	3.66 (3.37–4.01)	2.66	7	0.93	0	0	0

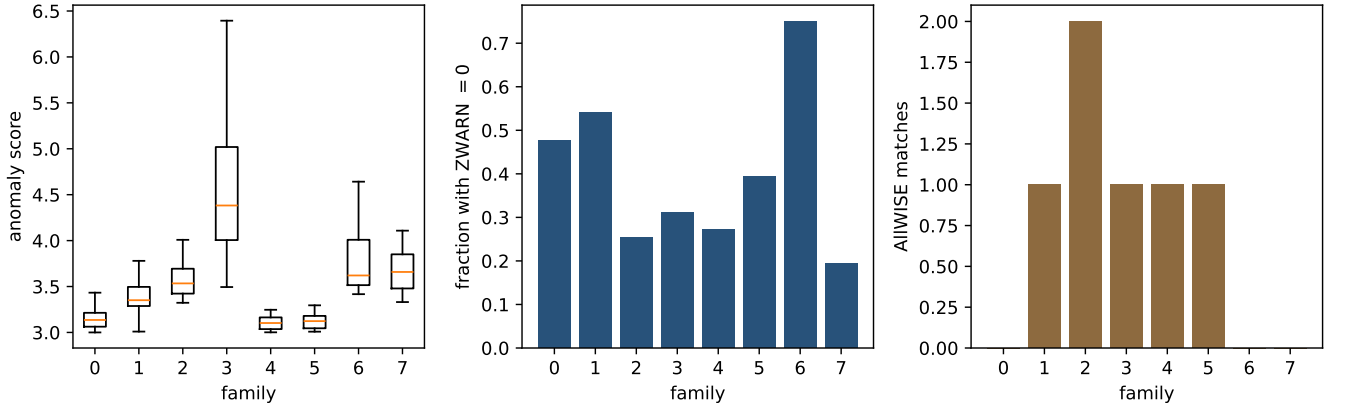


FIG. 7. Per-family anomaly-score distribution, reliable-redshift fraction, and AllWISE match count.

TABLE IX. Survey purity and sky extent of the eight families. Sky position enters the clustering only as a proxy for survey and programme footprint.

Family	N	Dom. share	R.A. span	Dec. span
0	302	78.5%	350.4°	94.7°
1	87	85.1%	260.7°	81.2°
2	71	46.5%	89.8°	67.3°
3	61	50.8%	147.1°	58.9°
4	44	65.9%	75.9°	40.5°
5	38	68.4%	72.4°	14.8°
6	36	75.0%	51.7°	23.3°
7	36	50.0%	114.7°	22.8°

AllWISE coverage of the unmatched subset is 6 of 675, so mid-infrared colours cannot discriminate between families at all; the 74 AllWISE matches in the full sample sit overwhelmingly in the matched half, which is expected, since catalogued objects are catalogued because they are detectable elsewhere.

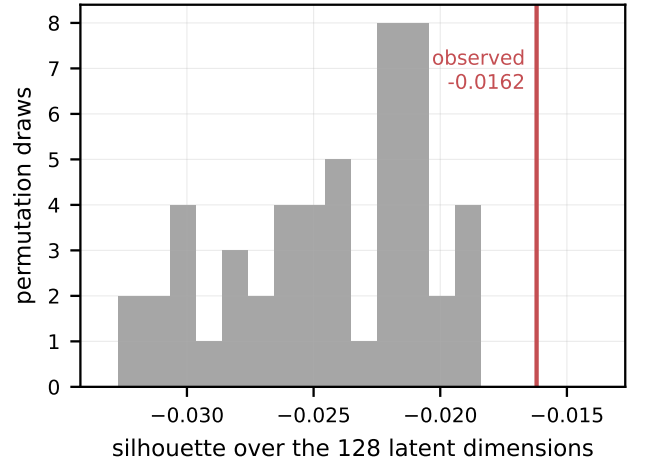


FIG. 8. Silhouette of the eight-family partition over the 128 latent dimensions, against its 50-draw label-permutation null. The observed value beats every draw but is itself negative: structured, but not separated.

VII. FOLLOW-UP SET AND OPEN TESTS

A. Named targets

Table X names 32 distinct objects across 35 tier rows in five rules-based tiers, all drawn from the unmatched subset. The rules were fixed from literature criteria rather than tuned to yield a target count, and are replayable from the released files.

- **FT-A** (4 targets) — $\text{ZWARN} = 0$ and $z \geq 4$: four anomaly-selected DESI-pipeline $z \geq 4$ quasar candidates with no catalogued counterpart, two of them at $z > 6$. These are pipeline redshifts, not confirmations. Section VII B runs the cheapest discriminating test on them here.
- **FT-B** (1 target) — AllWISE $W1 - W2 \geq 0.8$, the standard mid-infrared AGN criterion [11, 12]. The object is TARGETID 103610381762578 at $W1 - W2 = 0.822$, in family 5. Next tests are public archives: NEOWISE-R light curve [13], ZTF optical light curve [16], archival X-ray coverage.
- **FT-C** (5 targets) — $\text{ZWARN} = 0$, $\Delta\chi^2 > 25$ and S in the top 5%: spectra that are simultaneously well measured and hardest to reconstruct, where a genuine spectral anomaly is most likely. Scores run from 4.64 to 7.15 and three of the five are in family 3.
- **FT-D** (0 targets) — unmatched point sources with a Gaia DR3 counterpart [15]. The rule returns nothing, and that is a result: none of the 675 unmatched objects has a Gaia counterpart at all, so Galactic follow-up of this catalogue is not cheap. We retain the rule so a deeper generation can re-run it.
- **FT-E** (25 targets) — the highest-score member of each cluster, so that a follow-up programme tests the taxonomy rather than one corner of it.

B. A Lyman-break check on the FT-A candidates

The cheapest test in the whole follow-up set costs no observing time. A quasar at redshift z places Lyman- α at $1216(1+z)$ Å, and the intergalactic medium absorbs essentially everything blueward of it, so any broadband filter lying entirely below the break must go dark. For the four FT-A candidates the break falls at 6484, 7530, 8587 and 8616 Å, so the Legacy Survey g band (≈ 4000 – 5500 Å) is fully blueward of the break for all four, and the r band (≈ 5500 – 7000 Å) for the three at $z > 4.5$. The released photometry already contains the answer. Table XI and Fig. 9 give it.

Three outcomes, stated at their evidential strength:

Refuted.: TARGETID 39628216868014338 ($z = 5.194$) carries $f_g = 4.33$ nmgy, 15 times the catalogue’s median g flux and the 95th percentile of the whole sample,

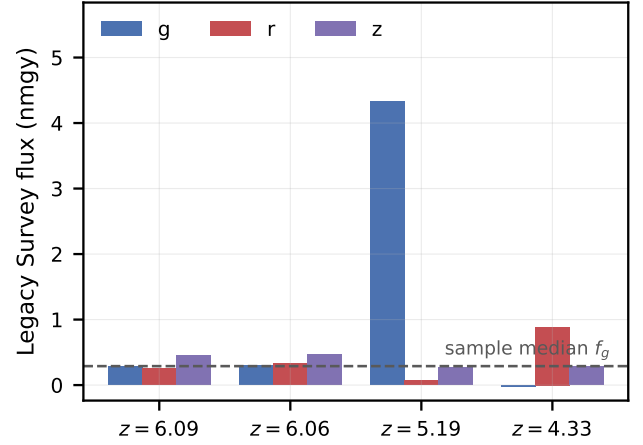


FIG. 9. Legacy Survey grz fluxes of the four FT-A candidates. A Lyman break at the pipeline redshift requires f_g to vanish in every panel. The $z = 5.19$ candidate instead carries the 95th-percentile g flux of the whole catalogue.

with $f_g/f_r = 59$. Both g and r lie fully blueward of a $z = 5.19$ break, so this flux pattern is inconsistent with the pipeline redshift under any plausible noise level. Its $\Delta\chi^2$ of 22.0 is also the weakest of the four, and its morphology is SER rather than point-like. We classify it as *disfavoured by public imaging* — either the photometry is blended or the redshift solution is wrong — and retain it in the table so that the tier rule can be audited.

Supported.: TARGETID 39627849358909321 ($z = 4.334$) carries $f_g = -0.019$ nmgy, consistent with zero and in the 0.8th percentile of the catalogue, with $f_r = 0.886$ nmgy — the break at 6484 Å sits inside the r band, so a surviving r flux is expected. Its $\Delta\chi^2$ is 4205, by far the strongest redshift fit in the tier. This candidate passes the cheap test and is the one worth spectroscopy.

Undecidable from the release.: The two $z \approx 6.07$ candidates carry $f_g = 0.283$ and 0.296 nmgy — 0.98 and 1.02 times the catalogue median, at the 48th and 52nd percentiles. Those values are exactly what a faint, undetected source looks like in this catalogue, but because the released schema omits flux inverse variances we cannot say whether they are detections or noise, and therefore cannot either confirm or refute a g dropout. Legacy Survey DR10 forced photometry with inverse variances settles both objects at zero observing cost, and is the immediate next step.

We emphasise what this does and does not establish. One candidate has been removed by public data alone, and one has cleared its first hurdle; neither outcome makes any object here a discovery, and a $\text{ZWARN} = 0$ Redrock solution on a low-signal-to-noise spectrum fails routinely at high redshift. The remaining tests, in cost order, are: visual inspection of the public DR1 coadd for a Lyman- α break and forest; DESI DR2 repeat coadds

TABLE X. Named follow-up targets. S is the anomaly score. Redshifts are DESI pipeline values and are printed only where $\text{ZWARN} = 0$; a dash marks an object whose fit carries $\text{ZWARN} = 4$ (`SMALL_DELTA_CHI2`), for which the pipeline redshift is not a measurement. The FT-E tier spans the taxonomy by construction and therefore includes such objects.

Tier	TARGETID	R.A.	Dec.	S	z	ZWARN	SPECTYPE
FT-A	39627903570285763	247.89529	4.66991	3.496	6.0878	0	QSO
FT-A	39627607293039577	201.10137	-7.55334	3.232	6.0638	0	QSO
FT-A	39628216868014338	239.92839	18.01577	3.061	5.1938	0	QSO
FT-A	39627849358909321	252.47160	2.60073	3.033	4.3338	0	QSO
FT-B	103610381762578	223.91434	50.21732	3.105	–	4	GALAXY
FT-C	39627812453222421	212.59529	0.92095	7.151	1.0798	0	GALAXY
FT-C	39627812461610773	213.07783	0.96584	6.394	1.1868	0	GALAXY
FT-C	39627877817258319	149.92219	3.77396	6.180	1.1989	0	GALAXY
FT-C	39633001079899014	217.98717	36.00150	4.747	1.3533	0	GALAXY
FT-C	39633407088529181	97.49930	61.65664	4.642	1.3098	0	GALAXY
FT-E	39627812453222421	212.59529	0.92095	7.151	1.0798	0	GALAXY
FT-E	39633407088529181	97.49930	61.65664	4.642	1.3098	0	GALAXY
FT-E	39636182677589446	64.05184	-16.74091	4.321	0.0063	0	GALAXY
FT-E	1071051379310604	240.34378	42.49037	4.289	–	4	GALAXY
FT-E	39627748041295714	333.31949	-1.73961	4.252	1.4203	0	GALAXY
FT-E	39628291237221824	260.29164	21.13162	4.108	–	4	GALAXY
FT-E	39633396858619848	275.81585	60.77425	3.877	–	4	QSO
FT-E	1070259272417281	210.20609	6.12193	3.762	–	4	GALAXY
FT-E	1070185960177669	150.13551	2.91550	3.498	–	4	GALAXY
FT-E	39627903570285763	247.89529	4.66991	3.496	6.0878	0	QSO
FT-E	67803973419009	130.11975	18.65811	3.491	–	4	GALAXY
FT-E	39633418778053415	143.59822	62.82238	3.416	1.3659	0	GALAXY
FT-E	39628330143583760	240.61454	22.94852	3.400	3.1050	0	QSO
FT-E	39633432057219468	96.54153	63.88652	3.387	0.9304	0	GALAXY
FT-E	39633495382820413	102.70638	70.64374	3.334	–	4	GALAXY
FT-E	39633466035275117	293.95821	67.33832	3.330	1.4545	0	GALAXY
FT-E	39627817989705740	182.69435	1.34328	3.325	1.2954	0	GALAXY
FT-E	2394988512018434	203.98955	49.98956	3.295	–	4	GALAXY
FT-E	39633062929108448	182.23513	39.17878	3.279	0.2303	0	GALAXY
FT-E	39632946235181990	213.68195	33.19121	3.278	–	4	GALAXY
FT-E	1070090426515458	215.89907	-1.05994	3.247	–	4	GALAXY
FT-E	39627778101872111	325.03381	-0.42236	3.220	1.0189	0	GALAXY
FT-E	2394624320602116	188.48854	30.29515	3.216	–	4	GALAXY
FT-E	39633184245159076	270.34970	45.93852	3.194	1.4010	0	GALAXY
FT-E	39627654445405293	147.36232	-5.59562	3.158	–	4	GALAXY

TABLE XI. The four FT-A candidates. Fluxes are Legacy Survey values in nanomaggies as carried in the released catalogue; the sample-wide median f_g is 0.289 nmgy.

TARGETID	z	S	$\Delta\chi^2$	f_g	f_r	f_z
39627903570285763	6.0878	3.496	23.4	0.283	0.255	0.446
39627607293039577	6.0638	3.232	135.5	0.296	0.330	0.470
39628216868014338	5.1938	3.061	22.0	4.326	0.073	0.262
39627849358909321	4.3338	3.033	4205.1	-0.019	0.886	0.286

where the tile is revisited; and only then confirmation spectroscopy, optical for the $z \approx 4$ candidate and near-infrared for the $z \approx 6$ pair.

C. Open tests

OT-1 Selection function.: No injection-recovery experiment has been run for this model on this substrate. The catalogue therefore has no measured completeness or purity, and no population-level inference may be drawn from it. This is the gate any rate, abundance, or cos-

mological use must clear first, and it requires injecting known spectral classes into DR1 coadds at controlled signal-to-noise and re-scoring with the hash-bound model — GPU-scale work, not local. **[PLACEHOLDER: run and report per-class injection recovery with the exact model and substrate named]**

OT-2 Latent-space taxonomy. Re-clustering on the 128 latent dimensions rather than on (S, α, δ) , following the negative silhouette of Sec. VIC. Cheap; the released catalogue already carries the latents.

OT-3 Blue-arm diagnosis. Compare the b -camera residual against airmass, lunar phase, and blue-end sky-subtraction residuals in the DR1 exposure tables, to separate calibration from astrophysics. This decides how much of the catalogue is astrophysical and is the highest-value open test after OT-1.

OT-4 Wider recovery benchmark. Six of eleven candidate reference classes lacked usable positions and were not tested; recovering their coordinates would materially strengthen or weaken Sec. VD.

OT-5 Deeper scan. A self-supervised model trained on the full science-target spectrum population, rather than a 47k-spectrum autoencoder, is the realistic route to a confirmed anomalous class.

VIII. WHAT THIS CATALOGUE BEARS ON

This catalogue was built inside a programme whose interest in DESI spectra is an early-universe one: assembling anomalies in existing public data that strain single-field inflation, and attaching an explicit discriminator to each. It is worth being precise about how little of that this release supports, because the temptation to over-connect a candidate list to a cosmological question is exactly what a provenance-first release exists to resist.

Three statements are defensible. First, the catalogue is *infrastructure* for such a programme rather than evidence within it: it supplies a reproducible, hash-bound path from a public survey to a candidate list, and a validation contract a third party can re-run. Second, the only channel by which this particular catalogue could feed an early-universe abundance argument is its high-redshift end — the 36 reliable-redshift objects at $z \geq 4$ and 16 at $z \geq 6$, of which the four unmatched FT-A candidates are the interesting subset. Third, that channel is closed at present, and for two independent reasons: the candidates are unconfirmed, and Sec. VIIB has already removed one of four on public data alone; and without OT-1 the catalogue has no selection function, so it admits no abundance, rate, or number-density statement of any kind.

We also record a null from the same programme that bounds what a catalogue like this could ever contribute. A local-shape primordial non-Gaussianity of the amplitude predicted by a matter-dominated contraction shifts

the $z \approx 10$ –12 massive-halo abundance by roughly 6–15%, more than an order of magnitude below the $\gtrsim 0.3$ dex stellar-mass systematic floor of current high-redshift photometric censuses, and with the wrong sign to explain an over-abundance; a factor-of-two effect would require $|f_{\text{NL}}| \sim 30$, far outside current constraints. The high-redshift abundance is therefore not where that class of model is measurable, whatever a candidate catalogue contributes to it. Publishing that null beside the catalogue is the point: it is what keeps the release from being read as evidence it is not.

IX. DATA AVAILABILITY AND REPRODUCIBILITY

The catalogue, its manifests, the validation checker and its output, the per-family evidence tables, the follow-up set, and the figure and table generators are released in the project repository under [results_2026-08-07/-phase3_v2](#) and [flagship_assembly_2026-09-18](#) (both under [pipelines/p1_highz_tracers/](#)), with mirrors on Hugging Face and Backblaze B2 verified by checksum. The full reproducibility manifest — external data sources and links, APIs, scripts, compute venue, cost, and wall-clock — is [REPRODUCIBILITY_MANIFEST.md](#). **[PLACEHOLDER: Zenodo DOI for the public deposit]**

Reproduction has two stages with very different costs.

Stage A — scan and characterisation. The scan, enrichment, cross-match, AllWISE query and taxonomy ran end to end on a single rented GPU pod at \$0.17/hr in 7 h 42 m 53 s, for approximately **\$1.31** of compute. The dominant stages are not compute-bound: enrichment took 2 h 43 m and the SIMBAD/NED cross-match 3 h 24 m, the latter limited by public service rate limits, so it reproduces at similar wall-clock on any machine. Only the scoring pass requires a GPU. All DESI, SIMBAD, NED, VizieR and Legacy Survey inputs are public and need no account.

Stage B — every number in this paper. The assembly reads only the ten hashed artifacts, uses no network, and runs on a laptop CPU in under 60 s at essentially zero cost:

```
python3 assemble_flagship_evidence.py
python3 make_tables.py
python3 make_draft_figures.py
```

It emits the validation-contract outcome, the family evidence, the follow-up set, every table under [tables/](#), every figure under [figures/](#), and [outputs/draft_numbers.json](#), which carries each derived quantity quoted in this text. Every number in this manuscript names its source artifact in a comment in the L^AT_EX source. The pipeline is deterministic apart from the V12 permutation null, which is seeded.

The archived autoencoder weights are a lab artifact, mirrored and hash-bound; retraining from scratch is not

part of this reproduction path, while scoring is fully reproducible given the frozen weights. The selection function does not exist for this model–substrate pair (OT-1), and nothing in this paper depends on it.

X. WHAT THIS CATALOGUE IS NOT

- It is not a discovery catalogue. No object here is reported as a confirmed unusual source; selection-pipeline output alone never promotes an object to notable.
- It is not a statistically characterised sample. Without OT-1 there is no completeness, no purity, and no selection function, so no population inference is available.
- Its families are not physical classes (Sec. VIC).
- Its unmatched fraction is not a novelty fraction: it includes objects simply outside other surveys’ footprints or below their detection limits, and the population is faint enough that only six objects in the whole catalogue have a Gaia counterpart.
- Its anomaly score is a distance from one model’s learned reconstruction, not a physically motivated significance, and Sec. VC shows it is dominated by one spectrograph arm.
- Its high-redshift candidates are pipeline redshifts, not confirmations — and one of four has already failed a public-data check.

XI. CONCLUSIONS

We publish 1,244 DESI DR1 science-target spectra selected by reconstruction-error anomaly score — the top 5.7×10^{-5} of the 21.8 million science targets scanned — with a sealed provenance chain, a published validation contract of sixteen machine-checkable requirements, a descriptive taxonomy of the unmatched subset, and a named follow-up set.

Four results are worth carrying forward independently of the catalogue. A score cut without a provenance gate selects sky fibres, not objects, and the contamination worsens with score; this is a property of the public coadd substrate and applies to any reconstruction-error search run on it. Within the science-target sample the score is not a proxy for exposure quality or brightness. The score is driven by the blue spectrograph arm, which localises both the next methodological test and the most plausible instrumental explanation. And the taxonomy’s families, tested in the model’s own latent space, are structured but not separated — descriptive strata, not classes.

The known-object recovery benchmark does not clear its pre-declared bar, and the one follow-up test cheap enough to run here removed one of four high-redshift candidates on public imaging alone. We therefore report the

work as what it is: reproducible data infrastructure with a validated selection, a stratified candidate list, and a named list of the tests it has not yet passed.

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This research has made use of the SIMBAD database [8], operated at CDS, Strasbourg, France; the NASA/IPAC Extragalactic Database (NED), which is funded by the National Aeronautics and Space Administration and operated by the California Institute of Technology; and the VizieR catalogue access tool, CDS, Strasbourg, France, all queried through astroquery [28].

All computation for this manuscript’s assembled numbers (Sec. A) ran locally on the author’s own hardware at zero incremental cost. The upstream scan and characterisation stage (Sec. 2) used a single low-cost RunPod GPU rental (\$1.31 total); no proprietary or allocated compute time was used.

AI-assisted methodology.—AI coding and review tools, operated under the author’s direction, assisted with implementation review, documentation, manuscript drafting and revision, and adversarial cross-checking and adversarial review. Their outputs were not treated as evidence: all catalogue assertions reported here were accepted only after direct checking against the machine-readable artifacts, committed code, and local or public-

source validation. The author set the study scope, made the scientific judgments, and takes full responsibility for the manuscript. No AI system is an author or is credited as one.

Appendix A: Software

The assembly uses NumPy [22], pandas [23], SciPy [24], scikit-learn [25], Matplotlib [26], Apache Arrow/PyArrow [27], and astroquery [28] for the SIMBAD, NED and VizieR queries. Exact versions are recorded in the reproducibility manifest.

The four citations not recorded in any committed benchmark artifact (Ref. [19]’s journal/volume/page, and the three software citations Refs. [25, 26, 28]) were verified against ADS for this manuscript (2026-09-19): Ritter & Kolb 2003, *Astron. Astrophys.* **404**, 301 (ADS bibcode 2003A&A...404..301R); Pedregosa *et al.* 2011, *J. Mach. Learn. Res.* **12**, 2825 (JMLR volume 12 official record); Hunter 2007, *Comput. Sci. Eng.* **9**, 90 (ADS bibcode 2007CSE.....9...90H); and Ginsburg *et al.* 2019, *Astron. J.* **157**, 98 (ADS bibcode 2019AJ....157...98G, arXiv:1901.04520). All four entries below already carried the correct values; no bibliography text changed.

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